

Abdul Rehman

Data Engineer | Software Engineer

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PROFESSIONAL SUMMARY

Detail-oriented Data Engineer with hands-on experience designing and building scalable ETL/ELT pipelines, data warehouses, and analytics platforms. Proficient in Python, PySpark, SQL, and cloud data services including Snowflake and GCP. Strong foundation in data modeling, pipeline orchestration, and ensuring data quality at scale. Exposure to OpenAI automation using agentic workflows.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, PySpark), SQL, Shell Scripting (Bash), REST API Development

Data Engineering: ETL/ELT Pipeline Development, Data Warehousing, Snowflake, Data Quality Frameworks.

Big Data & Processing: Apache Spark/PySpark (DataFrames, RDDs, Spark SQL), Apache Airflow, Apache Kafka, Hadoop Ecosystem (HDFS, Hive), Snowflake, Power BI, Batch processing.

Cloud Platforms: Snowflake (Warehouses, Stages, Pipes, Tasks), GCP (BigQuery, Dataflow, Cloud Storage, Pub/Sub, Cloud Functions)

Databases: PostgreSQL, MySQL, MongoDB, Redis.

DevOps & Tools: Docker, Git, CI/CD Platforms GitHub Actions, Terraform, Linux.

Security & Compliance: Data Breach Remediation, Data Governance, Access Control

ML/AI: OpenAI, Hugging Face, n8n

PROFESSIONAL EXPERIENCE

Data Engineer

January 2025 – Present

Accounting Solutions, Greensboro, NC.

- Designed and implemented ETL pipelines using Python and PySpark to ingest, transform, and load 50K+ records daily from multiple sources into Snowflake data warehouse.
- Built data models using star and snowflake schema designs, optimizing query performance by 40% through proper indexing and partitioning strategies.
- Developed Apache Airflow DAGs to orchestrate complex data workflows, implementing error handling, retry logic, and alerting mechanisms.
- Created data quality validation frameworks using Great Expectations to ensure pipeline reliability and data integrity across transformations.
- Implemented incremental data loading patterns and CDC (Change Data Capture) mechanisms to optimize pipeline efficiency and reduce processing time.
- Created drill-down reports and blended data from multiple sources (Snowflake, SQL Server, flat files) into unified Tableau visualizations.
- Deployed self hosted n8n, and created an OpenAI agent workflow to update the Google worksheets based on the CDC data.

Junior Data Engineer

December 2023 – November 2024

MS Associates LLC - Remote.

- Designed and implemented enterprise SQL Server data warehouse using Medallion Architecture (Bronze, Silver, Gold layers), enabling scalable data integration from ERP and CRM source systems with full data lineage tracking.
- Built automated alerting mechanisms for pipeline failures, SLA breaches, and data quality anomalies using email notifications and integration with incident management systems.
- Created data validation checkpoints across Bronze-to-Silver and Silver-to-Gold transitions, ensuring data integrity through schema validation, null checks, and business rule enforcement.
- Developed SQL-based analytics, KPI dashboards, and ad-hoc reports enabling data-driven decision making for business stakeholders across sales, finance, and operations.
- Delivered end-to-end BI solutions using Tableau and Power BI, translating raw pipeline data into actionable dashboards for executive and operational teams.

EDUCATION

Bachelor of Software Engineering

University of Agriculture, Faisalabad

September 2020 – May 2024